MOBILE APPLICATION DEVELOPMENT LAB

Course Code:	**CS3604-1**	**Course Type:**	**PCC Lab**
Teaching Hours/Week (L: T: P):	**0:0:2**	**Credits:**	**01**
Total Teaching Hours:	**0+0+26**	**CIE + SEE Marks:**	**50+50**

Teaching Department: Computer Science & Engineering

Course Objectives:

1.	**Describe** the architecture and overview of android.
2.	**Develop** a mobile application on android platform using UI components and Android Components.
3.	**Demonstrate** data handling in Android Develop a mobile application on android platform using SQLite
4.	**Build** an Android web service.
5.	**Develop** application to demonstrate google map and navigation.

List of Experiments

PART-A

1.	Design four checkboxes namely any four food items and one button. Find total amount of food items selected in Toast message after clicking the button.
2.	Design simple calculator application that performs basic arithmetic operations. Use ADD, SUB, MUL, DIV buttons to perform operations, CLEAR button to reset the fields, and edit text widgets for reading operands, displaying result value.
3.	Create an application which generates a random color on each click.
4.	Implement the options menu concept in the application to choose between two activities (Give appropriate titles to activities).
5.	Implement context menu concept in application to change the background color.
6.	Design an application to send SMS using Intent class.
7.	Design a phone call application that takes a phone number from the user.
8.	Write an application to toast your joining date and course selected for engineering using a Date picker and List view/Spinner.
9.	Write an application to make a dialogue box to confirm the change of background color or image.
10.	Design an application that captures the image using a camera and set the captured image as the background for your application.

PART-B

1.	Implement a service concept to play the music in the background for long duration and perform a foreground job.
2.	Implement an AsyncTask to count from 1 to 1000 in the background and the display the progress using progress bar on the screen.
3.	Implement broadcast receiver to carry out the of following: Read battery charge of your mobile, display it using progress bar and change the background color as given in table. **Color** / **Battery Charge** Red — 0% To 20% Blue — 21% To 60% Green — 61% To 100%
4.	Write an application to insert the data entered by a user into a database and display all the

	values in database.
5.	Write an application to search for a given USN from a student database and call to that student.
6.	Design a simple resume builder application using two activities that take basic information namely Name, Email Id, Mobile No, Gender, Qualification, and Profile Picture from the user. Use appropriate text widgets for (Name, Email Id, Mobile No, Qualification), radio buttons for gender, Imageview for the profile picture.
7.	Write an application that creates a notification message that will launch another activity after clicking on it.
8.	Implement web view concept in application that contains two activities and opens default web page/user entered web page.
9.	Implement an application to store and retrieve data by using shared preference. (Include save, delete and retrieve operations)
10.	Implement the following animation concept i. Blink ii. Move the image object iii. Rotate. iv. Zoom In and Out

Course Outcomes: At the end of the course student will be able to

1.	Familiarize with the Android platform and Android Studio environment, designing user interfaces using Android UI components and application components.
2.	Apply concepts such as SQLite, shared preferences, files, broadcasts, notifications, sensor APIs, and telephony APIs to develop robust and functional Android applications.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→ / **↓ Course Outcomes**	1	2	3	4	5	6	7	8	9	10	11	12	**PSO↓** 1	2	3
CS3604-1.1	3	3	3	-	1	1	-	2	2	1	2	1	2	3	-
CS3604-1.2	3	3	3	-	1	1	-	2	2	1	2	1	2	3	-

1: Low 2: Medium 3: High

REFERENCE BOOKS:

1.	Anubhav Paradhan, Anil V Deshpande, "Mobile apps Development",First Edition, Wiley, 2014.
2.	Barry Burd , "Android Application Development All in one for Dummies", Second Edition Wiley, 2015.
3.	SAMS,"Teach Yourself Android Application Development in 24 Hours",FirstEdition,Sams Publishing,2010.
4.	Wei-Meng Lee, "Beginning Android Application Development", Wrox Publication, 2011.
5.	Reto Meier, "Professional Android 4 Application Development", Wrox Publication, 2012..

E Resources

1.	https://www.tutorialspoint.com/android/index.htm
2.	https://www.javatpoint.com/android-tutorial
3.	https://developer.android.com/guide/
4.	http://nptel.ac.in/courses/106106156/
5.	https://www.youtube.com/watch?v=SYoNOvdZ3M&list=PLonJJ3BVjZW6CtAMbJz1XD8ELUs1KXaTD&index=19